Decision Records Under Uncertainty: Correspondence, Rule Sensitivity, and Added Information

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Abstract

External decision records can reproduce a model's action while leaving their diagnostic value unresolved. We examine record availability, correspondence, rule sensitivity and added information in three predecessor panels and a 48-assignment follow-up separating record requirements from external execution. The follow-up has 46 completed trials and two retained technical losses. When records were required but model proposals were executed directly, Astra and Sol each produced eight valid, proposal-matching records; both supported rules selected the same action in every such record. Sol's eight matching records included two conditional loss labels. Thus observed correspondence coexisted with rule invariance and did not ensure continuation feasibility. In the external-executor condition all completed proposals already matched the selector, leaving correction of conflicting proposals unmeasured. All follow-up P/L changes occurred in two continuous-control histories; the purposeful histories and repeats support descriptive comparisons only. The predecessor transfer adapter supported no complete pairs. In its separate 32-trial common-menu successor, native actions and rule identity determined all 32 observed-rule cells. Native actions plus a single-distinct-row check determined 29 of 32 supported-rule-family cells; a full per-record scan determined all 32 by construction of the contract. One pilot conclusion changed under a native-compatible selector completion. These cases show why an analyst must separate availability, correspondence and sensitivity before attributing information to a crossing. Task labels reuse a bound menu census and retain numerical/runtime premises; full task success, internal faithfulness and improvement in independently scored diagnosis are unmeasured.

1 Question and contribution

Studying how a model acts under incomplete knowledge requires connecting what it records to what it chooses. Our broader research objective is to understand how models formulate assumptions, select informative experiments and revise predictions. Before using external records as evidence about that process, we ask what those records actually establish: are they available, do their declared rules reproduce the proposal, and does substituting a rule or pairing records expose a distinction unavailable from simpler checks? These are measurement questions. The present experiments do not test whether a model's chosen experiments resolve more uncertainty or improve held-out predictions.

We study external decision records consisting of a representation and a rule. In the common-menu experiment, a representation is a numeric table of failure forecasts and a rule selects a column. These numbers are uncalibrated forecasts; they are not verified world models or a report of all internal reasoning. An analyst has privileged access to the task evaluator. A resulting action is labeled preservation (P), loss (L), or unresolved (U) according to the stated remaining-task feasibility procedure. None of these labels identifies a model's internal cause merely because it is attached to an executable record.

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The diagnostic proposal examined here originated in our own component pilot: cross a recorded representation with another recorded rule, then use the resulting action's task label to characterize their contribution. It is not presented as an established procedure in widespread use. Three concrete observations constrain that proposal. First, reproducing a native action did not uniquely fix the analyst's selector completion: five native-compatible completions gave different crossed outcomes. Second, optional records were scarce under the frozen adapter, whereas requiring records exposed same-rule and row degeneracies. Third, the follow-up's required-record/model-proposal condition produced eight matching records per profile without externally selecting the action, yet every such record was action-invariant under the two supported rules. Sol's matching records also included two conditional L outcomes. Correspondence, rule sensitivity and task feasibility therefore need separate readouts in this case.

The practical decision is whether a researcher has enough evidence to interpret a crossing as a diagnostic contrast, or must first test record availability, selector specification and within-record sensitivity. For example, our pilot's apparent off-diagonal P outcome becomes L under another native-compatible completion (Section 4.1); calling that a unique component diagnosis would hide an analyst choice. In the later common-menu panel, the sole P/L-changing rule contrast is already visible without a partner record. These examples locate unsupported inferences in this study; we have not measured whether applying the checks improves other researchers' diagnosis accuracy or future-behavior prediction.

The contribution is the empirical separation of these measurements and an executable reader that reconstructs them from identity-bound sources. The full-scan identity is a supporting structural control, not an empirical discovery. We do not propose a general attribution algorithm, establish that crossing is useless across contracts, or demonstrate internal faithfulness.

2 What the comparison measures

2.1 An external computation and two different crossings

Let A_i be record i's table over a common finite menu and let r_i be its declared deterministic rule. Native action $a_i = r_i(A_i)$ is one column of that menu. The *observed-rule crossing* has four entries $c_{ij} = r_j(A_i)$ for $i, j \in \{1, 2\}$. The separate *supported-rule-family table* evaluates each A_i under both allowed rules: uniform mean and worst case, with lowest-index tie breaking. The family table may contain a rule that neither paired actor used. It must not be described as a second live observation or an observed model counterfactual. Task labels for these actions come from the bound menu census.

2.2 Same-access baselines and a structural identity

The comparison holds the archived tables, rule definitions, menu and task-label map fixed. It removes operations rather than withholding information from a baseline. We evaluate (i) native actions and rule equality, (ii) native actions and the single-distinct-row condition, and (iii) independent per-record evaluation of each supported rule. The last baseline never needs a partner's table. Only after producing its results do we compare them with paired cells.

Proposition (per-record sufficiency under a fixed contract). For fixed tables A_i, a common menu, and a fixed family $\mathcal{R}$ of deterministic rules with specified tie breaking, the collection $\{(i, r, r(A_i)) : r \in \mathcal{R}\}$ determines every supported-rule-family cell and every observed-rule cell whose rule belongs to $\mathcal{R}$. *Proof.* Each cell is one of those evaluations; no partner table is an argument of $r(A_i)$. A fixed action-label map also determines its label. This statement does not establish the validity of that map or identify an actor's internal computation.

Two weaker checks need fewer rule evaluations. If $r_1 = r_2$, then $c_{i1} = c_{i2} = a_i$. If a table has a single distinct row, mean and worst case select the same column. Duplicate-containing tables with other distinct rows do not satisfy this condition. The full-scan equality is thus a compositional identity of the contract. The empirical question is how often the weaker checks suffice and what contrasts remain; agreement with

the full scan is not an independent discovery of the identity. We measure recoverable cells, not a wall-clock speedup or causal attribution accuracy.

Task and record at a glance. In the continuous task, an actor sees a state/action history and chooses a bounded two-dimensional control; the remaining objective is to reach and maintain the target region without violating a path constraint. In the route task, it sees a discrete history and chooses a legal route action under the remaining task constraints. In this panel each task supplies a finite action menu and permits up to three analyses of the observed history. A decision record is a table: columns are menu actions, rows are uncalibrated candidate failure forecasts in $[0, 1]$, and the declared rule minimizes the mean or the worst row value. In the older 32-trial panel, the external selector executes that choice in its mandatory condition. Appendix G specifies the exact histories, menus and six-field schema.

For a didactic two-action table $A = \left(\begin{smallmatrix} 0 & 0.4 \\ 0.7 & 0.4 \end{smallmatrix}\right)$, mean scores $(0.35, 0.4)$ select action 0, whereas worst-case scores $(0.7, 0.4)$ select action 1. This invented arithmetic example is not a trial or a task-label claim; it shows what a per-record scan can reveal.

2.3 Feasibility labels and access

P requires a successful admissible continuation witness after the action. L requires the specified loss certificate; absence of a witness alone is not L. U preserves unresolved numerical cases. Continuous-control labels are conditional on the disclosed floating-point and bounding premises in the supplement. The discrete route family's finite computation has a different verification contract. Both actors and analyst see the legal histories; only the analyst receives the hidden mechanisms and privileged evaluator during the original experiment. Reviewer disclosure now permits inspection of these mechanisms without retroactively changing what participants received.

Record validity, correspondence and rule sensitivity can be checked without the physical label map. P/L/U comparisons add its assumptions. The exact arithmetic helper does not prove the uniform library bounds or attest the historical runtime; those premises remain open. These are distinct evidence levels when interpreting outcome differences.

3 Evidence sets and their denominators

The first three evidence sets below use crossing-pair denominators. The separate follow-up in Section 5 instead compares A/B/C conditions within eight matched triplets per profile; these condition pairs are not crossing pairs. Its 48 assignments include two technical losses.

Evidence set	Trials	Pairs	Eligible records	Eligible pairs
Pilot (component study)	8	4	2 selected component pairs	
Prospective transfer	48	24	1	0
Optional record	16	8	1	0
Mandatory record + execution	16	8	16	8

Table 1: The three predecessor evidence sets. Pilot component eligibility is a different construction from the frozen prospective adapter and common-menu contract. Empty pair denominators leave the crossing effect unassessed.

The pilot used four fixed histories with two repeats under the requested gpt-6-astra/max profile. Its component study focused on two boundary histories, B and D, after outcomes were available. The transfer panel used 48 live trials: two families, three instances per family, two histories per instance, the gpt-6-astra/max and gpt-6.1-sol/max profiles, and two repeats. Under its frozen adapter, a record had to support the prescribed operation without post hoc completion. Only one trial qualified, leaving zero of 24 pairs crossable. This is an applicability result, not evidence that models lack unrecorded representations or that crossing has zero diagnostic effect. Native transfer labels were 42 P and 6 L. The adapter's mutually exclusive outcomes

were: 24 route trials with no analysis attempt, 18 whose last successful analysis lacked a table, three whose rule did not reproduce the action, two with a menu-identity mismatch, and one supported record. These reasons distinguish absent evidence from an invalid submitted record. The other 24, continuous-task trials made 72 analysis attempts: 67 succeeded and five returned errors. Thus not calling an analysis and calling one that fails are different observations; neither establishes absence of internal reasoning or inability to analyze.

The 32-trial panel crossed two task families, two purposefully selected histories per family, two model profiles (gpt-6-astra/max and gpt-6.1-sol/max), two repeats and two conditions. Each condition has 16 trials and eight pairs. A crossing pair comprises the two repeats at the same history and profile within one condition, not a pair of trials from different conditions. Both conditions used a common finite menu and structured final output. Appendix G identifies the selected histories, execution contract, resource limits and recorded usage. The optional condition permitted an observational record; the mandatory condition required a record and let an external deterministic executor select the action. Mandatory record–action agreement is consequently a construction property, unlike a matching optional observational record. The combined intervention does not isolate a pure reporting effect.

All 32 trials were evaluated: no unstarted slots, technical missingness or protocol-invalid trials remain. This does not mean the analysis interface was error-free: 48 attempts yielded 41 successes and seven code errors in seven trials (five optional, two mandatory). Five errors used attribute access on the dictionary history; two used the unavailable built-in `next`. The prompt described `history.states`/`history.actions`, although the worker passed a dictionary, and did not enumerate the built-ins. Each affected trial consumed all three analysis slots, one on the error and two successfully. Appendix I gives the trial-level accounting. No trial is removed or relabeled. The effect on subsequent actions and record structure is unassessed. Optional P/L/U counts were 13/3/0; mandatory counts were 10/5/1. Four purposefully chosen histories do not form a population sample of tasks or support a model ranking. Table 2 separates the families' label contracts. The study inventory in Appendix H identifies excluded historical and development work; the later acquisition experiment has its own denominator.

Condition	Family	P	L	U	Verification contract
Optional	Continuous	5	3	0	Numerical, conditional
Mandatory	Continuous	2	5	1	Numerical, conditional
Optional	Route	8	0	0	Finite integer recurrence
Mandatory	Route	8	0	0	Finite integer recurrence

Table 2: Native outcomes in the common-menu panel, eight trials per row. Continuous P uses an admissible continuation witness; continuous L depends on the numerical and historical-runtime premises in Appendix F. U remains unresolved. Successful present-host replay does not close those premises.

4 Results: applicability is not incremental information

The mandatory condition made all eight pairs executable. It did not make their crossing tables informative by construction. Every pair used the same declared rule, so all 32 observed-rule cells were already determined by native actions and rule identity. Among the 16 mandatory records, 13 had one row, one contained duplicate rows alongside distinct rows, and 13 had only one distinct row. The last count must not be increased by treating every duplicate as a degenerate table.

Fifteen of 16 mandatory records selected the same action under both supported rules. The remaining record changed action and its conditional continuous P/L category; the independent scan recovered this change. One pair involved U and remains unresolved rather than being converted into a resolved loss. The two 32-cell displays overlap; after within-pair deduplication they reference 30 unique numerical computations, not 64 new samples. Cross-action labels reuse the existing bound menu census. No new live model counterfactuals were collected for this retrospective analysis.

Baseline information	Observed cells	Family cells
Native actions + identical observed rules	32/32	—
Native actions + one-distinct-row check	—	29/32
Independent two-rule scan per record	32/32	32/32
Additional cells from full table after scan	0	0

Table 3: Mandatory condition only. Counts describe recoverable displayed cells, not independent observations or accuracy on external diagnoses.

Worked same-record contrast. For the mandatory `probe-01` record from gpt-6-astra, repeat 1, the independent scan yields:

Rule applied to this record	Menu index	Bound task label
Uniform mean	7	L (conditional continuous certificate)
Worst case	12	P (continuation witness)

Indices are zero-based. The recorded table, menu identity and label census are held fixed. No partner record is needed to expose this sensitivity; it concerns the declared external rule, not a demonstrated internal cause. The alternative action's label is a deterministic menu evaluation, not an additional live model decision. This is the sole rule-sensitive mandatory record, so it must not be presented as a prevalent population effect. It is also one of the trials affected by the dictionary-access error. The archived table's rule sensitivity is reproducible; whether correcting the interface would have changed that table or the decision is unknown.

The optional condition contains one submitted record, which is eligible, and 15 trials with no submitted record (`no_record`). It has no eligible pairs. Thus 15 absent records must not be described as 15 failed record implementations. Its incremental effect is unassessed, not zero. Thus this experiment shows both an applicability bottleneck without mandatory execution and a redundancy bottleneck after enforcing it. These are distinct conclusions.

4.1 Pilot contrast and selector sensitivity

Each selected pilot history already had one native P and one native L outcome. The declared B completion produced two P off-diagonals; D produced two L off-diagonals. These patterns distinguish conditional component dependencies but introduce no outcome category absent from the native pair. They are not an independently labeled ground truth for internal model causes.

In B, the off-diagonal operation required an analyst-specified completion. All five examined completions reproduced the native action, but selecting the plan with the smallest predicted trajectory peak changed the crossed outcome from P to L. Appendix C.1 defines all five selectors and reports their outcomes. The conclusion must therefore name the completion it uses. D's off-diagonal losses show a conditional joint-change pattern, not that each native decision was a failure. The pilot's heterogeneous optimizers do not implement the later common-menu comparison; its full-crossing versus per-record-scan result was not measured. We retain this distinction instead of extending the later structural identity to an untested setting.

5 Follow-up: required records with and without external execution

The earlier mandatory condition combined a record requirement with an external executor. The follow-up adds a required-record condition that executes the model's proposal. Each matched triplet uses one history, repeat and model/max/Fast profile: A permits an optional record and executes the proposal; B requires a valid record and executes the proposal; C requires a valid record and executes its declared selector, retaining the proposal for observation. Every trial starts a fresh context. Four purposefully selected historical contexts, two profiles and two repeats give 48 planned assignments, eight triplets per profile. These repeats are not independent draws of new worlds. The allocation and six possible within-triplet orders were fixed before

collection; earlier-panel trials are not reused as controls. Appendix J gives the execution segments, interface and reader boundaries. No new population sample or live crossed decision was collected.

Profile	Arm	Done	Tech.	P/L/U	Valid	Match	One row	Same action
Astra	A	7	1	6/1/0	0	–	0	0
Astra	B	8	0	8/0/0	8	8/8	8	8
Astra	C	7	1	5/2/0	7	7/7	6	6
Sol	A	8	0	7/1/0	0	–	0	0
Sol	B	8	0	6/2/0	8	8/8	6	8
Sol	C	8	0	7/1/0	8	8/8	7	7

Table 4: Follow-up, eight planned assignments per row; profiles are separate strata, not ranked. Done means a completed trial, not full task success. Valid counts usable records; Match is record-selected action versus model proposal among valid records. The empty A denominator is undefined. Same action counts records for which both supported rules select the same action. Sol C has one duplicate-containing record; all other rows have zero. There are no protocol losses, active or unstarted assignments.

Availability and correspondence. In A, no valid record was available among Astra's seven or Sol's eight completed trials. This is absence of the specified external record, not absence of reasoning. In B, each profile supplied eight valid records and all eight matched its own proposal. Unlike C's executed-action agreement, this correspondence was not imposed by the executor. It does not establish that the record caused the proposal or faithfully represents internal reasoning. All eight B records per profile were invariant in action across the two supported rules. Thus the earlier degeneracy observation also occurs without external action selection. Single-row and duplicate records were permitted; their presence is a scientific observation, not a protocol violation.

Conditional outcomes and observed opportunities. Table 5 reports the two predeclared contrasts using only observed nontechnical pairs. Astra has seven fully completed triplets and Sol eight. L outcomes remain in the comparison; technical failures are reported separately and are not converted into L. The directions do not support a common improvement claim across these profiles, nor an equivalence or no-effect conclusion. In Sol B, eight matching records coexist with two L labels, illustrating why correspondence and continuation feasibility require separate measurements. P certifies a conditional feasible continuation; the experiments do not measure completed-task success.

Profile	Contrast	Pairs	Missing	L→P	P→L	Same	Net P
Astra	B–A	7/8	1	1	0	6	+1
Astra	C–B	7/8	1	0	2	5	-2
Sol	B–A	8/8	0	0	1	7	-1
Sol	C–B	8/8	0	2	1	5	+1

Table 5: Descriptive paired transitions run from A to B for B–A, and from B to C for C–B. Pairs are observed/planned; Missing counts technical missing pairs; Same includes P-to-P and L-to-L. Dividing Net P by observed pairs recovers the reported paired difference. No pooled estimate or population inference. C–B is the total condition contrast, not an isolated executor effect on the same fixed record.

Where the outcome contrasts occur. The family breakdown in Table 6 locates every observed P/L movement in the two continuous-control histories. All completed route decisions were P, although each route menu contained an L option; this pattern was not forced by having only P choices. The continuous menus have 13 actions and the route menus three, so these purposeful contexts are not a calibrated difficulty ladder. Appendix J.1 reports every profile/history/arm cell. In particular, Sol's C–B net gain of one P comprises two L-to-P and one P-to-L transitions. Net change alone does not describe those opposing movements, and the underlying continuous labels retain the numerical/runtime premises stated above.

Profile	Family	Assigned	Done	Tech.	P/L/U
Astra	Continuous	12	12	0	9/3/0
Astra	Route	12	10	2	10/0/0
Sol	Continuous	12	12	0	8/4/0
Sol	Route	12	12	0	12/0/0

Table 6: Follow-up outcomes by family, separately within each profile. Each family has two purposeful histories and two repeats per arm. Done denotes a completed decision, not full task success. P/L labels in the continuous family retain the numerical/runtime premises.

A changed action need not change the task label. The follow-up contains two rule-sensitive records, both in C (Table 7). Mean and worst case select different menu indices, while the bound label remains P in one and L in the other. These are existing-record calculations, not new live counterfactual trials. Together with the older panel's single label-changing example, they make action sensitivity and task-label sensitivity separately inspectable.

Profile	History	Repeat	Arm	Mean	Worst case
Astra	probe-01	2	C	1 (P)	0 (P)
Sol	probe-02	1	C	11 (L)	0 (L)

Table 7: The two rule-sensitive follow-up records. Entries give zero-based menu index and bound label. The action changes but the label does not; these are per-record calculations on existing menus, not live crossed decisions. Continuous labels remain conditional.

Every completed C proposal already matched the record-selected action (Astra 7/7, Sol 8/8). No observed C trial therefore tests the benefit of overriding a conflicting proposal. The C–B contrast is not an isolated mechanical correction effect on the same fixed record. Labels and within-record rule diagnostics are tied to the existing menu contract; they are not additional live crossing observations or independent evidence about new task physics.

6 Position relative to existing work

CAR estimates effects of interventions on agent steps through same-policy downstream replay and also studies interactions (Shah, 2026). Our object is a declared external record; a crossed action with a privileged feasibility label is not a replay estimate of the stochastic policy's effect. AgentRx uses constraint evidence and a judge to diagnose execution traces (Barke et al., 2026). AGENTSCOPE combines execution dependencies with neural invariants for localization (Bi et al., 2026). We do not evaluate their failure-location accuracy or claim that structured evidence is new.

BLADE is a particularly relevant comparator: its dependency-guided handoff audits separate information use from the final attribution, and its ablations compare richer evidence with simpler configurations and selective with exhaustive inspection (Guo et al., 2026). Consequently, neither the general warning that structure alone is insufficient nor the value of a simpler baseline is our novelty claim. Our specific contribution is the decomposition of this crossing contract's outputs into applicability, same-rule identity, row degeneracy and remaining per-record rule sensitivity.

Faithfulness requires evaluating the relation between an explanation and the system it purports to explain (Jacovi & Goldberg, 2020). Interchange interventions on actual neural representations test a different relation (Geiger et al., 2021). Our record substitutions neither intervene on those representations nor establish that the external record generated the model's internal decision. The literature comparison is scoped to inspected primary sources; it is not an exhaustive novelty proof or a head-to-head experiment.

Behavior-linked explanation evaluation. Madsen et al. (2024) test self-explanation consistency against model behavior using counterfactual, attribution and redaction tasks. Mayne et al. (2026) measure whether

adding an explanation helps a predictor anticipate a reference model's answers on related inputs, relative to the same information without the explanation. Our record check instead recomputes a declared action from an archived table. It observes correspondence on that decision; it does not measure either behavior under the explanation's proposed input changes or incremental predictive information on new decisions. Those targets provide concrete extensions, not results established by our matching-record counts.

Connection to scientific discovery. BoxingGym evaluates experiment selection through expected information gain and model discovery through predictions, including whether a scientist's explanation helps another agent predict without further experiments (Gandhi et al., 2025). Our study examines how to interpret a decision record within such a broader research agenda; it does not benchmark those discovery abilities or outperform BoxingGym. The distinction is between a record-level measurement and independently assessed experimental or predictive utility. A correct external calculation alone cannot supply the latter.

7 Interpretation and remaining questions

The case supports a sequence of measurement decisions. An absent record leaves the proposed operation unsupported. A native-compatible record may still leave selector completion ambiguous. A matching proposal may be action-invariant under the supported rules and may receive an L label. Even a rule-sensitive action may preserve its task label. Each observation answers a different question; none by itself establishes an internal cause or a gain in diagnostic accuracy.

For researchers using this contract, these distinctions determine which additional analysis is justified. The transfer panel's empty pair denominator calls for an applicability assessment before interpreting crossing effects. The pilot requires reporting selector-completion sensitivity. In the common-menu panel, a per-record scan already exposes the remaining rule sensitivity; the paired display adds no independently evaluated diagnostic target. The follow-up shows that matching records and action invariance can coexist without enforced action selection. It does not establish that these checks improve real-world diagnosis or that a different contract is redundant.

The broader research program remains open. Three next questions connect records to behavior: whether a stated assumption predicts the next experiment choice; whether that choice reduces more uncertainty than a valid fixed plan with the same resources; and whether a revised explanation predicts cases withheld during its construction. Each needs an independently scored outcome and a design that gives the behavior a real opportunity to occur. The present purposeful histories, repeated contexts and fixed two-rule family do not answer those questions. More trials under the identical crossing contract would not turn its full-scan identity into evidence of added diagnostic value.

Several boundaries also remain empirical. B correspondence does not identify whether the record caused the proposal. All completed C proposals already matched their selectors, leaving mechanical correction of conflicting proposals unmeasured. The seven analysis errors in the older 32-trial panel have an unisolated effect; their occurrence includes its sole label-changing rule-sensitive record. A later corrected interface does not establish what that earlier record would have been. Continuous task labels retain their numerical/runtime premises, whereas record-level calculations do not require those physical claims. P is a continuation witness, not measured full-task completion. No population effect, profile ranking, calibrated difficulty scale or internal reasoning faithfulness follows from this study.

An optional companion note addresses one bounded part of the broader program: selecting an additional observation for target certification in a finite class of deterministic systems. Its 5,000 paired draws yield a 1.78 percentage-point gain with interval [1.23, 2.29] points. It uses deterministic policies and additional local computation, not live LLM discovery. The note's methods and compute costs are separate supporting material; its result is not needed to establish the record findings above and is not pooled with them.

8 Reader access and reproducibility

The accompanying review bundle contains the LaTeX sources, disclosed mechanisms, pilot/transfer numerical replay and the 32-trial record comparison. It also regenerates all 32 menu labels from identity-bound histories and mechanisms. Exact condition instructions, schema, all 80 archived request payloads and a renderer recovering their original prompt hashes are included; archived participant analysis text is never executed.

The companion's offline path reconstructs the original catalog and exclusions, seeded rejection draw, 5,000 eight-step histories, both policies' queries and certificates, target audits, counts, work and primary interval. Original committed record bytes and frozen sources are checked before comparison. This deterministic replay of existing inputs is distinct from collecting new observations or reattesting historical processes and elapsed times.

The follow-up supplies all 46 completed minimal final JSON responses, including the 31 numeric decision tables and their declared rules, the original allocation, and the common and A/B/C instructions with their wire schema. The standard-library reader derives record validity, declared-rule selection, both supported rule actions and row diagnostics directly from the preserved numeric lexemes. It compares the derived decisions to the archived results, verifies menu and profile bindings, and checks all 46 completed labels by lookup in the existing bound menu census. This is a source-table and label-link check, not fresh physical grading or independent historical attestation. Participant analysis programs, intermediate analysis outputs and raw trajectories are not included. The older scientific inputs remain unchanged; this revision did not rerun their physics.

From the bundle root, run:

```
python3.12 -I -B -S verify_all.py --followup-only
```

This checks the exact package, source-derived follow-up decisions, family/history counts, directional paired changes and the generated manuscript tables. Omitting the flag also runs the unchanged older numerical and acquisition checks; the README describes their previously verified scope and current validation boundary. No model account or service is needed. Same-host or clean-directory replay is not independent laboratory replication. Historical evidence remains preserved; this preparation is not a claim of public release, submission or acceptance.

Responsible interpretation. The experiments use simulated environments and model outputs, not human-subject data. Agreement with an executable record is not evidence of honest intentions, internal reasoning faithfulness, or safe deployment. These distinctions matter if the checks are reused to assess consequential agent decisions. The disclosed mechanisms support inspection of this fixed study, not a claim that a future benchmark remains secret or resists contamination.

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A Disclosed task, mechanism and delivered information

Represent the two-dimensional state and action by complex numbers z_t and u_t. A step is indexed by its pre-action state: u_0 produces z_1. The initial conditions are $z_0 = 1$, $\phi_0 = 4.147589101011308$ and $u_{-1} = 0$. The evaluated world obeys

$$d(z) = (1.06 + 0.015|z|^2)z + (0.12 - 0.08i), \tag{1}$$

$$z_{t+1} = d(z_t) + e^{i\phi_t}u_t, \qquad |u_t| \le 0.6, \tag{2}$$

$$\phi_{t+1} = \phi_t + \omega + \kappa \,\mathrm{Re}(u_{t-1})\,\mathrm{Im}(u_{t-1}), \qquad \omega = 0.1. \tag{3}$$

The lag in the phase update is part of the mechanism: the update uses u_{t-1}, not the action just chosen. The original family allowed $\kappa \in \{0.3, 3, 6, 12, 18\}$. Only the two settings below occur in the fixed-history pilot. They are parameter settings, not a validated difficulty ordering.

Paper alias	Archival case identifier	Snapshot after step	κ
A	case-01	4	12
B	boundary-01	11	12
C	case-03	4	18
D	boundary-02	9	18

An episode ends on escape ($|z_t| > 4$) or after 32 actions. Success requires completion and $|z_t| \le 0.05 + 10^{-10}$ at *all* $t \in \{28, 29, 30, 31, 32\}$. Merely avoiding escape is insufficient. Action validation rejects nonfinite coordinates or norms above $0.6 + 10^{-9}$; a smaller tolerance overshoot is normalized to 0.6. State observations are rounded to twelve decimal places, whereas the simulator retains binary floating-point state. Stored actions retain their applied precision. The exact numeric history of every selected case is included in `legacy-replay/data/cases.json`; it is not regenerated from a caption or rounded manuscript table.

The participant receives the objective, horizon, action limit, escape threshold, full permitted action–state history and remaining analysis allowance. It does not receive these equations, ϕ_t, ω, κ, the evaluator's continuation or its loss diagnosis. The opaque identifiers in delivered inputs do not disclose the boundary/comparison labels. The frozen selector and schemas are inspectable in the reader artifact; the full provider logs are not redistributed. Disclosure to reviewers does not alter the historical participant information boundary.

B Witness and loss-bound justification

Positive witness. For every pre-action snapshot and every resulting state, one fixed privileged controller attempts a continuation. At each step it chooses

$$u = \mathrm{clip}_{0.6}\left[-d(z)e^{-i\phi}\right], \qquad \mathrm{clip}_r(v) = \begin{cases} v, & \|v\| \le r, \\ \dfrac{rv}{\|v\|}, & \|v\| > r. \end{cases} \tag{4}$$

Here $\|v\|$ is the Euclidean norm, equal to the modulus for a complex scalar. The original transition, action validation, escape check and scheduled target test score the entire continuation. A successful admissible continuation establishes existence for the implemented task. A failed continuation does not establish impossibility. This controller knows the hidden phase and mechanism; its success is not evidence of blind action identification or participant learning. Every pilot pre-action state passed this positive baseline.

Sufficient negative bound for the ideal recurrence. Write $x = (\mathrm{Re}\, z, \mathrm{Im}\, z)$ and $c = (0.12, -0.08)$. Fix a vector q with $\|q\|_2 \le 1$. Suppose $q^T x \ge p > 0$. Since $\|x\|_2 \ge q^T x \ge p$, and the scalar multiplier $1.06 + 0.015\|x\|_2^2$ is positive, the next-state projection satisfies

$$q^T x' = (1.06 + 0.015\|x\|_2^2)q^T x + q^T c + q^T R(\phi)u \tag{5}$$

$$\ge (1.06 + 0.015p^2)p + q^T c - 0.6. \tag{6}$$

The last term follows from Cauchy–Schwarz and the admissible action radius. The function $g(p) = (1.06 + 0.015p^2)p$ is increasing on the positive domain. Consequently, iterating this lower bound while it remains positive gives a bound for *every* allowed future control sequence, irrespective of the future phases. A bound exceeding 4 before the horizon implies unavoidable escape; a bound exceeding the target tolerance at a required target step excludes successful completion. A target violation already present in the recorded prefix also suffices, because future actions cannot change a past required state.

Implementation and numerical scope. The evaluator chooses q by truncating each coordinate of $x/\|x\|$ toward zero to four decimal places, and checks its norm. It initializes $p_0 = q^T x$ from the stored full-precision state. The implemented recurrence is

$$p_{j+1} = \left\lfloor (1.06 + 0.015p_j^2)p_j + q^T c - 0.6 - 10^{-9} \right\rfloor_{10^{-12}}, \tag{7}$$

where $\lfloor \cdot \rfloor_{10^{-12}}$ rounds downward to that grid. Coefficients and initial state are converted from their actual binary-float values to exact rational numbers. Loss is reported only for a positive induction sequence crossing 4.000001 within the remaining horizon, or crossing $0.05+10^{-10}$ at a required target step. A separate 110-digit decimal implementation checks the direction, every bound, the positivity domain, threshold and horizon. The code also checks that the transition definition and task constants match the bound's assumptions.

Archived arithmetic audit. A subsequent record-only audit covers the pilot, distinct component grades and the continuous subset of the prospective panel. All 50 distinct stored directions, including directions for which no loss was certified, have exact squared norm at most one; the minimum value of $1-\|q\|_2^2$ is 5.75×10^{-6}. There are ten distinct certified projection records (eleven occurrences). Rational recomputation reproduces the archived binary64 bounds exactly. Recomputing these recurrences with per-step subtractions 10^{-9}, 2×10^{-9} and 10^{-8} preserves every certificate kind and first certification step. Even at the enlarged slacks, the smallest final lower-bound gap above the applicable runtime threshold exceeds 0.0371. No simulator transition or model request is performed by this audit, and no historical certificate is rewritten.

The action validator rejects computed norms above $0.6 + 10^{-9}$ and rescales a smaller computed overshoot by $a\,0.6/|a|$. It therefore does not simply apply every action up to the rejection tolerance unchanged. For the 1,201 distinct applied actions in the inspected continuous trajectory records, exact squared-norm checks give a maximum norm of approximately 0.60000000000000002694. This finite record check does not bound all possible future actions or transcendental-function errors.

Runtime-specific conditional budget. A separate local audit inspects the pinned interpreter and the currently loaded ARM64 math image. Its compiled complex multiplication uses fused multiply-add instructions; its complex exponential calls `exp` and `__sincos_stret`. The norm uses platform `hypot`, and the square still uses platform `pow`. Thus a source-level list of separate sine/cosine calls does not fully describe this executable.

Appendix F derives a norm bound for the captured `hypot` instruction sequence under ordinary IEEE binary64 round-to-nearest semantics with gradual underflow. It then reduces the remaining library accuracy requirements to an absolute square error ϵ_{pow} and a norm excess δ_{rot} for the computed complex rotation. Accuracy relative to the true angle is not needed for this projection bound. Taking each of these two bounds as 2^{-30} yields total one-step error below 6.288×10^{-10}, which fits the archived 10^{-9} subtraction. The local checker verifies the scalar inequalities with exact rational arithmetic.

This is a sufficient conditional bound, not an attestation that the two library bounds hold uniformly. Finite current-process probes satisfy them but do not prove them. In addition, the two inspected historical runtime inventories and activation records do not pin the loaded math-image identity or floating-point control register. Today's image capture cannot retroactively fill that gap. The ten archived projection certificates therefore retain their stated conditional numerical status. The route family's exact integer transitions and finite-horizon dynamic program do not use this projection premise.

The classifier returns preserved only with a successful witness, lost only with a sufficient negative certificate, and unresolved otherwise. Simultaneous positive and negative evidence is treated as an error. This is an

incomplete classifier, not a complete viability kernel or a map of a continuous safe cone. A numerical failure in candidate optimization is a different unresolved category and is not fed to this classifier as physical loss.

C Computational interventions and reproducible selectors

The candidate coefficients and original analysis outputs are fixed data. No candidate is refitted after a component is crossed. Let $f_m + B_m u$ denote candidate m's predicted next state at the selected snapshot. Candidate construction, feature families, phase extrapolation and the exact numeric coefficients are inspectable in `legacy-replay/data/component_inputs.json`. The replay implements these fixed operations; it does not execute the archived participant programs.

One-step worst-case rule. For G equally spaced angles and radius r, set $u_j = r(\cos(2\pi j/G), \sin(2\pi j/G))$ and choose the first grid index minimizing $\max_m \|f_m + B_m u_j\|_2$. Snapshot B uses $G = 720$ and $r = 0.6$; D uses $G = 1440$ and $r = 0.599999999$. The four finer-grid checks use $G = 2880$. This is optimization on a finite circle, not a guarantee of global optimization over the full action disk or of task completion.

Past-error weighted rule. Each fitted parent has weight $w_m = 1/(10^{-4} + e_m^2)$, where e_m is its recorded leave-one-out residual RMS. In the expanded D component, six scenarios inherit equal shares $w_m/6$ of each parent's weight. With $v = \sum_m w_m B_m^T f_m$, the selected action is $u = -0.6v/\|v\|$. The matrices here are scalar gains times rotations, so the quadratic term in $\sum_m w_m \|f_m + B_m u\|^2$ is constant on the radius-0.6 circle. Thus this direction minimizes that fixed-radius surrogate for these matrices. It is not a general theorem for arbitrary linear-control matrices, a calibrated posterior, or a worst-case task-safety guarantee. D's second representation comprises 11 parent fits expanded by two phase modes and three phase offsets $\{-0.12, 0, 0.12\}$; the complete construction is in the code and numeric inputs.

Multi-step plans and the researcher completion. For each of B's candidates, the individual planner optimizes the remaining $N = 32 - n$ controls with projection to the action disk. Its surrogate objective is

$$J_m = \sum_{i=1}^{N} \left[w_{n+i} \|\hat{x}_{n+i}\|^2 + \lambda \big(\max\{0, \|\hat{x}_{n+i}\|^2 - 14.44\} \big)^2 + 10^{-5} \|u_{n+i-1}\|^2 \right], \tag{8}$$

where $w_t = 1$ for $t \geq 28$ and 10^{-4} otherwise, and $\lambda = 10$. The deterministic initialization, projected accelerated-gradient updates, backtracking limits and restart schedule are included in `components.py`; individual plans use 1,000 iterations. The declared selector R2* chooses the individual plan with the lowest computed cost and takes its first action. This final selection rule was completed by the researchers. Agreement with the original choice is a reconstruction check, not proof that it was the model's latent policy. At least one crossed action has magnitude approximately 0.5755, so this contrast does not isolate direction or horizon while holding magnitude fixed.

The two auxiliary joint plans use a shared first action, separate candidate-specific later actions, the mean of the candidate costs, and 1,400 iterations. One is numerically unresolved. The one-at-a-time D deletions remove each of 66 scenarios or each of 11 six-scenario parent groups; all other settings stay fixed and no refitting occurs. No-rescue results cover only these interventions, not joint deletion, changed fitting or arbitrary alternative uncertainty models.

The 91 planned cells comprise 8 primary crossings, 4 finer-grid checks, 2 joint-plan auxiliaries and 77 omissions. Ninety were computed and one auxiliary remained numerically unresolved. Repeated actions are deduplicated before privileged grading, giving 13 distinct grades. There is no additional model sample in this count. The full records, rather than the rounded values in figures, define action identity.

C.1 Five retrospective completions at B

This separate sensitivity analysis holds the eight individual plans from B's native-loss pool A1 and four from its native-preservation pool A2 fixed. It changes only which plan supplies the first action. For a plan's predicted states $\hat{x}_t$, the five minimized criteria are: the total penalized cost J_m above; $\max_{28 \leq t \leq 32} \|\hat{x}_t\|$;

$\frac{1}{5}\sum_{t=28}^{32}\|\hat{x}_t\|^2$; $\max_{n\le t\le 32}\|\hat{x}_t\|$, including the common starting state; and $\|\hat{x}_{32}\|$. All ties use the lowest existing plan index. The last criterion covers only the final target step; it is not the full task-success condition. The finite selector list was fixed before its additional grading, with the original outcomes already known.

Criterion minimized	Native A2	Crossed A1	A1 action norm
Total penalized cost (original R2*)	P	P	0.575465
Maximum target-step norm	P	P	0.575465
Mean squared target-step norm	P	P	0.575465
Trajectory peak norm	P	L	0.042684
Final-step norm	P	P	0.579114

Table 8: B selector sensitivity from the archived fixed-plan analysis. All five reproduce the native A2 action within 10^{-6}; the first three choose the same A1 plan. The minimum-peak completion reverses the reported rescue. Selector counts are not probabilities or independent replications.

Thus native-action reproduction alone does not identify the crossed completion. The reversal also changes the chosen plan and action magnitude; it does not isolate a horizon-only or direction-only effect. The saved selector indices, exact actions, metrics and labels are in `artifact/source/pilot-sensitivity.private.json`. This finite sensitivity result does not cover new optimizers, mixtures or arbitrary alternative policies.

D Protocol, record boundaries and cost accounting

The fixed trial order was A1, D1, C1, B1, C2, B2, A2, D2, where the suffix denotes repeat. Each trial begins in a fresh local context with an opaque snapshot identifier; both repeats of a case receive exactly the same initial payload bytes. Each subsequent payload contains the same physical history and that trial's own submitted analysis code and returned results. Therefore the third or fourth requests are not equal-input repetitions of the first request.

The wire schema permits one analysis request or one final two-coordinate action. Analysis code must define `act(history)` and can use the allowed mathematical facilities; no network, filesystem, process creation or hidden simulator interface is provided. The protocol and policy source state the exact permitted syntax and names. Local analysis has 30-second CPU/wall limits, a sampled 1 GiB RSS stop and a 100,000-character output limit. Sampled RSS is not a hard allocation ceiling. Each trial permits at most three analysis requests and one final action, for at most four model requests in total. Trial wall time is 3,780 seconds; request time is bounded by the smaller of 900 seconds and remaining trial time minus a 120-second record/grading margin. No retry, neutral request, replacement or outcome-dependent stopping is permitted.

The model identifier and effort are configuration records, not attested server weights. Local receipts record one agent message per call, no unexpected tool events and cleanup of initially empty call directories. These checks do not establish independent sampling seeds or provider-side isolation. Full provider transcripts remain in the source archive; this bundle distributes selected numerical inputs and final record projections. Hidden chain-of-thought is not observed or claimed as evidence.

The recorded 588,051 input and 129,677 output tokens total 717,728. The 92,160 cached-input tokens are already included in input. Reasoning-output tokens are likewise a subset of output. There were 28 receipted requests and zero unreceipted requests in this pilot. Provider receipts are the source of these counts; analyst use and money are unmeasured. The historical technical null's unresolved reservation is not a zero-use observation and is not included in the pilot total.

E Prospective tasks, reference histories and adapter

Exact configurations and twelve packets are in `legacy-replay/data/transfer/`; the numeric replay inputs under `legacy-replay/data/transfer/` are checked against these records.

Continuous reference histories. The recurrence and task target are those in Appendix A, with three new initial phase/frequency configurations and couplings 3, 6 and 12. Phases and frequencies derive deterministically from SHA-256 of the literal seed `transfer-validation-20261025-v1` and instance ordinal. A prescribed privileged ten-step path tracks waypoints with radii $1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.75, 1.9, 2.05, 2.2$ and deterministic changing angles. Snapshots are after steps 6 and 10. Pre-collection checks require legal histories and a positive continuation, without resampling. This selection does not establish calibrated boundary distances or natural participant behavior.

Route–resource dynamics and certificates. The state is node $n \in \{0, \dots, 18\}$ and remaining integer fuel f. Action $j \in \{0, 1, 2\}$ is encoded as $[j, 0]$ and advances to $n' = \min(18, n+j+1)$. The hidden stationary positive edge cost is

$$c(n, j) = b_j + ((n + o_j) \bmod p_j), \qquad f' = f - c(n, j). \tag{9}$$

Success means reaching node 18 within twelve actions without negative fuel. The three fixed (b, o, p) configurations and initial fuel values are in `CASES.private.json`. Unlike continuous control, every legal route action is in the menu. The participant knows the graph, objective, horizon and observed costs through its action–state history, but not the edge-cost formula or private coefficients.

Let $V_h(n)$ be the minimum fuel needed to reach the goal in at most h further actions. Set $V_0(18) = 0$, $V_0(n < 18) = +\infty$, and

$$V_h(18) = 0, \qquad V_h(n) = \min_{j \in \{0,1,2\}} \{c(n, j) + V_{h-1}(\min(18, n + j + 1))\}. \tag{10}$$

The code represents infinity by `None`. With positive costs, $f \geq V_h(n)$ provides a minimum-cost admissible continuation; $f < V_h(n)$, negative fuel or an unreachable goal certifies loss. Every value-table cell is checked by a separately implemented recursive recurrence. Positive witnesses are replayed. Initial fuel is the exact minimum completion cost plus one; the lexicographically first minimum-cost path supplies snapshots after steps 2 and 4. Baseline feasibility does not prove that the hidden cost law is identifiable from the observed short prefix.

Exact optional table contract. The diagnostic continuous menu has twelve equally spaced directions at radius 0.6, rounded to twelve decimals, plus zero. For a returned table with losses L_{mj} and weights w_m, the weighted rule chooses the first menu index minimizing $\sum_m w_m L_{mj} / \sum_m w_m$; the worst-case rule minimizes $\max_m L_{mj}$ with the same tie break. Weights belong to the candidate pool. The adapter accepts only the exact keys `actions`, `candidate_losses`, `weights`, and `rule`, the exact delivered menu, 1–128 candidate rows and finite entries of absolute magnitude at most 10^{100}. Weights are finite, nonnegative and bounded by 10^{100}, with positive total. Final-action reproduction uses maximum coordinate error $\leq 10^{-8}$. The last successful analysis is selected before examining outcome category. A successful earlier table cannot rescue a missing or unfaithful final table.

The replay changes only imports/data paths, preserves frozen numeric functions, and omits dispatch code. Original function text and source/export digests are included. It reads bound archived analysis values; participant programs remain unexecuted text. It reproduces the eligibility test without inferring or regenerating tables.

F Narrow runtime proof audit

This appendix states a sufficient numerical contract for the continuous projection argument and documents which parts were established locally. The real-arithmetic argument, the archived recurrence computation, the current machine implementation and the historical execution environment are distinct evidence layers. None of the checks below adds a live agent trial or reruns a physical episode.

Identity and trust assumptions. The inspected executable matches the experimental Python digest. It is CPython 3.12.14 on ARM64. The current math image is `libsystem_m.dylib`; its UUID, text/constants hashes and relevant function bytes were recorded. The current thread reports a zero floating-point control

register: round to nearest, with no flush-to-zero or alternative handling enabled. We assume the ordinary IEEE binary64 semantics of the decoded add, multiply, divide, fused multiply-add and square-root instructions. The source-to-instruction interpretation is a manual derivation with exact constant checks, not a proof-assistant-certified binary analysis. The historical inventories pin Python but do not attest the loaded math image or control register; equivalence to the historical environment remains an explicit assumption.

Bound for the captured norm implementation. Let $u = 2^{-53}$ and $d = 2^{-1075}$, the unit roundoff and half the smallest subnormal. Here u denotes roundoff, not a physical action. For finite binary64 coordinates, put $a = \max(|x_1|, |x_2|)$ and $b = \min(|x_1|, |x_2|)$, and let $r = \sqrt{a^2 + b^2}$ in exact arithmetic. The captured function returns a when $b = 0$. Otherwise it drops b only if the stored exponent difference exceeds 27. This implies $b/a < 2^{-27}$, including a subnormal b, so dropping it reduces the exact norm by less than $r2^{-55} = ru/4$.

For nonzero retained inputs, the decoded branches use the following scales, where e_a is the biased exponent field of a:

Exponent field	Input scale	Undo scale
$e_a < 539$	2^{768}	2^{-768}
$539 \le e_a < 1534$	1	1
$e_a \ge 1534$	2^{-768}	2^{768}

The retained inputs scale exactly. In the small-input branch even the smallest subnormal becomes 2^{-306}. In the large-input branch a nonzero retained b has exponent difference at most 27 and remains normal after scaling. In all branches the squared larger input and the sum are normal and cannot overflow. The core computes $t = \mathrm{RN}(a_s^2)$, $v = \mathrm{RN}(b_s^2 + t)$ by a fused operation, and $w = \mathrm{RN}(\sqrt{v})$, before undoing the scale. Thus w lies within factors $(1-u)^2$ and $(1+u)^2$ of the scaled norm. Undoing a power-of-two scale is exact for a finite normal output; subnormal rounding contributes at most d. Overflowed outputs are not finite accepted norms. Combining the discarded-input allowance gives the sufficient bound, for a finite returned norm h,

$$r(1-3u) - d \;\le\; h \;\le\; r(1+3u) + d. \tag{11}$$

The rational checker verifies $(1-u/4)(1-u)^2 \ge 1-3u$ and $(1+u)^2 \le 1+3u$. In particular, $h \le 4$ implies $r < 5$ and $|h - r| < H = 16u$. This is an analytic bound for the captured sequence under the stated instruction semantics, not an accuracy inference from test samples.

Action normalization. Let L be the exact binary64 value of 0.6. An accepted but unscaled action satisfies $h \le L$, hence $r \le (L+d)/(1-3u) < L + 8u$. When $h > L$, the validator rescales by aL/h. Here $r > 1/2$ and $h > 1/2$. The compiled multiplication by a real scalar and division by a positive real denominator each round each component once; the zero imaginary terms do not introduce a separate nonzero error. Using $\sqrt{2} < 3/2$ gives

$$\|a_{\mathrm{applied}}\|_2 \le \frac{L(1+u)^2}{1-3u-2d} + 3d(1+u) + \tfrac{3}{2}d < L + 8u. \tag{12}$$

This covers accepted rescaling, including subnormal allowances, rather than treating the rejection tolerance as an applied-action allowance.

Two remaining library accuracy contracts. Write $\widehat{s} = \mathrm{pow}(h, 2)$ and $C(\phi) = \mathrm{cmath.exp}(i\phi)$ for the computed values. The needed contracts are

$$|\widehat{s} - h^2| \le \epsilon_{\mathrm{pow}}, \qquad \|C(\phi)\|_2 \le 1 + \delta_{\mathrm{rot}}. \tag{13}$$

They must hold uniformly over relevant norm inputs and permitted finite phases. The second includes the wrapper's `exp` and combined `__sincos_stret` computation. No bound on angular error is needed: every such control vector has norm at most $(1+\delta_{\mathrm{rot}})(L+8u)$, irrespective of its direction. Inspection of call targets establishes the operation path but does not prove either accuracy contract.

Composed one-step bound. For a state before computed escape, $r < 5$. Let α, β, c, L denote the exact binary64 task constants, so the reference drift is $(\alpha + \beta r^2)x + c$. Set $B = 32u + d$. Under the sufficient choices $\epsilon_{\mathrm{pow}} = \delta_{\mathrm{rot}} = 2^{-30}$, the relevant basic-operation intermediates have magnitude below 32; the square is handled separately by its library contract. Each basic rounding error is therefore at most B. The squared-radius error is bounded by $10H + H^2 + \epsilon_{\mathrm{pow}}$. Two basic operations form the drift multiplier. Scaling the state and adding the bias give vector error at most $5\beta(10H + H^2 + \epsilon_{\mathrm{pow}}) + 13B$. The compiled complex product uses one ordinary product and one fused operation per component; its error contributes less than $3B$. Final vector addition contributes less than $3B/2$.

Consequently the sufficient loss allowance beyond the ideal $-L$ term is

$$E = 5\beta(10H + H^2 + \epsilon_{\mathrm{pow}}) + 18B + 8u + \delta_{\mathrm{rot}}(L + 8u). \tag{14}$$

The exact rational evaluation is approximately $6.2870891 \times 10^{-10} < 10^{-9}$. For $\|q\|_2 \le 1$ and a positive projection lower bound p, this yields

$$q^\top x' \ge (\alpha + \beta p^2)p + q^\top c - L - E.$$

The archived subtraction and downward floor therefore remain sufficient *if* the stated instruction, library and historical-equivalence premises hold. The prior audit's minimum terminal gap exceeds 0.0371; it also clears the norm-test error. The same ten unique archived projection certificates are covered conditionally, with no relabeling.

Checks and open obligations. The archived local audit checked exact inequality constants, captured function identity and historical inventory coverage. The reader bundle rechecks only the exact constants and exponent ranges with `artifact/numeric_constants.py`; it does not recapture or attest the historical runtime. Fifty-two norm inputs, including exponent/underflow edges, twenty square inputs and twenty-six phases were probed in the archived audit interpreter. Every finite tested output satisfies the sufficient bounds; an overflowed norm is recorded separately. These are scalar diagnostics, not uniform accuracy proofs. Negative controls reject altered function bytes and oversized error budgets, and retain an open status when a library premise or historical binding is missing. They test the checker, not the missing premises themselves.

The remaining obligations are explicit: prove both bounds in Eq. (13) for the relevant library implementation, and justify its equivalence and floating-point settings in the historical runs. Current image hashes cannot supply absent historical attestation. Accordingly, this addendum narrows the numerical assumptions and supports their composition; it does not turn the archived results into unconditional machine-verified claims. Exact route dynamic programming and the observed trial counts are unchanged.

G The 32-trial common-menu protocol and panel usage

Selected histories and menus. Table 9 identifies the four purposeful histories using the earlier transfer panel's source IDs. The two continuous histories are at physical step 10; the two route histories are at step 2. Each appears with both profiles, both repeats and both conditions. These are selected reference histories, not a random sample of histories or eight newly sampled worlds.

Delivered ID	Source ID	Family	Instance	Cutoff	Menu size
probe-01	snapshot-03	Continuous	1	10	13
probe-02	snapshot-07	Continuous	2	10	13
probe-03	snapshot-02	Route	1	2	3
probe-04	snapshot-06	Route	2	2	3

Table 9: Fixed histories for the 32-trial panel. Source histories and menu identities are hash-bound in the archived allocation.

Both conditions receive identical history and task/menu bytes for a given probe; the condition instruction differs. The continuous menu is the twelve radius-0.6 directions at angles $2\pi j/12$, $j = 0, \ldots, 11$, rounded

to twelve decimals, followed by zero. The route menu is $[(0,0),(1,0),(2,0)]$ in that order. Only the indexed menu action is executed. The original physical clock, remaining horizon and task objective are unchanged. The participant receives visible history, public task/menu and its own analysis outputs, without hidden dynamics or future observations. Trial order was fixed before the run and is distributed in `artifact/data/crossing.json`.

Common output scaffold and condition-specific execution. The shared structured-output scaffold wraps a `decision` object that can request analysis or return an action ID, a record, or both. It does not itself enforce condition-specific fields, record identity, dimensions, loss range or action agreement; the local contract does. A record has exactly six fields: `schema`, `case_id`, `menu_sha256`, `loss_definition_id`, `candidate_losses` and `rule`. It contains 1–128 rows, one numeric loss in $[0,1]$ per menu action, and no weights. Each row uses the same declared meaning: an uncalibrated forecast of failing the original full task after that next action and the best continuation the participant can plan. The two rules minimize the uniform row mean or row maximum. Decimal numeric tokens are evaluated as exact rationals; ties select the lowest menu index. Single and duplicate rows are legal, and duplicates retain multiplicity. Rows are not verified world models.

In the optional condition, a legal `action_id` is required and an accompanying record is optional. Record failure or disagreement is logged without replacing that action. In the mandatory condition, only the record is allowed; its declared rule deterministically selects the executed action. An invalid record produces no action and no retry. Thus mandatory agreement is by construction, while optional agreement is an observational check. This six-field contract differs from the 48-trial panel's weighted, four-field last-analysis adapter in Appendix E; their eligibility counts are not measurements of the same interface.

Budgets and recorded use. For the 32-trial panel, each fresh trial permits up to three optional local analyses and one final decision: at most four model requests, 3,780 seconds per trial and 900 seconds per request. Analysis has a 30-second CPU/wall limit, sampled 1 GiB RSS stop and 100,000-character JSON output cap. It yields no new physical observation. No retry, repair prompt, neutral request or outcome-dependent extension is allowed; the panel ceiling is 128 model requests. The completed panel used 48 analysis requests and 32 final decisions. Its single selected action per trial was graded; crossed labels reuse the bound menu census, not additional live observations.

Panel	Requests	Input	Output	Cached input	Unknown calls
48-trial transfer	120	2,325,280	484,134	372,352	0
32-trial common menu	80	1,733,978	422,952	263,168	0

Table 10: Separate participant-model usage ledgers. Cached input is already included in input, and reasoning tokens, when reported, are part of output. Neither category is added a second time. Pilot usage is in Appendix D.

Table 10 reports archived receipt-derived counts, not analyst usage or provider charges; these latter quantities were not measured. Synthetic transport and privileged local computation are outside the table. The 32-trial plan reserves 1,050,000 input plus 128,000 output tokens for each request lacking usage, separately from measured totals. Its final unknown-call count is zero, so no such reserve remains in that panel's final accounting; this policy is not a provider billing cap. The source-bound scalar projection is `artifact/data/protocol-summary.json`. It provides provenance and protocol identities without redistributing the full provider logs.

Exact requests and executable reader path. `artifact/protocol/` contains both condition instructions, the shared wire schema, the four initial packets and all 80 request payloads, including each trial's accumulated analysis records. Its checker renders every full prompt and verifies the archived prompt and payload hashes. For example,

```
python3.12 -I -B -S artifact/protocol/check.py --show-call 1
```

prints the first request. The historical collection entry point is supplied as inspection-only text; provider transport credentials and full provider responses are not redistributed. Configured profile names are not independent attestation of server weights.

`artifact/menu_replay.py` binds each public packet to its transfer history and menu, then recomputes the 26 continuous and six route menu labels. The original record comparison and this numerical replay are separate checks. Continuous loss labels retain the open uniform-library and historical-runtime premises in Appendix F; present-host agreement does not discharge them.

H Study inventory and historical boundaries

The eight-trial pilot, 48-trial prospective transfer panel, 32-trial common-menu panel and separate 48-assignment follow-up are four evidence sets in the main paper. The two 48-slot studies are distinct. Their counts are never pooled. The follow-up has 46 completed trials and two technical losses, not 48 completed scientific trials. The two 1,600-draw development samples and the 5,000-draw main acquisition study appear only in the companion; the earlier prediction criteria failed and are retained there. The acquisition result is not an additional live LLM or crossing experiment.

Older archive labels describe different work: *H4* denotes an unfinished collection whose 84 remaining assignments stayed paused and whose crossing status is unassessed due to missing evidence. The *historical technical null* denotes an earlier infrastructure-limited record, not a successful zero-use scientific trial; its unresolved usage reservation stays outside the pilot accounting. These records supply no extra evaluable trials here. The archive's v1.8/v1.9 labels identify preserved predecessor research versions, not additional studies or independent replications. Current paper revision numbers track presentation and reader-access changes. This revision additionally incorporates the already completed follow-up as a separate evidence set; predecessor scientific inputs, denominators and the primary acquisition procedure remain unchanged.

I Analysis-interface errors and consumed slots

The historical shared instruction describes `history.states` and `history.actions`; the evaluator actually supplies a Python dictionary. Five analysis calls failed on `history.states`. Two other calls failed because `next` is absent from the restricted built-ins; the historical instruction refers to the existing interface without listing those built-ins. These are observed interface-use errors, not missing final decisions. The seven affected trials are listed below. Each made three analysis attempts: one failed and two succeeded. Errors consumed an attempt and were visible to the next request. They did not trigger replacement trials or extra slots.

History	Profile	Condition	Repeat	Error
probe-01	gpt-6-astra	Mandatory	1	Dictionary attribute
probe-01	gpt-6-astra	Optional	2	Dictionary attribute
probe-01	gpt-6.1-sol	Mandatory	2	Dictionary attribute
probe-01	gpt-6.1-sol	Optional	1	Unavailable `next`
probe-01	gpt-6.1-sol	Optional	2	Unavailable `next`
probe-02	gpt-6-astra	Optional	1	Dictionary attribute
probe-02	gpt-6.1-sol	Optional	2	Dictionary attribute

Table 11: All seven affected common-menu trials. All are continuous-task trials; both profiles requested max reasoning effort. The first row is the sole rule-sensitive mandatory record. These are descriptive associations, not estimates of the errors' effects on actions or records.

The 32 common-menu trials produced 48 analyses: 41 successful and seven erroneous. In the separate transfer panel, all 24 route trials made zero analysis attempts, while the 24 continuous trials made 72 attempts: 67 successful and five erroneous. These denominators are not pooled. The frozen adapter still supports only one transfer record and no complete pair.

`artifact/interface_audit.py` reconstructs all 32 trial rows from the last cumulative analysis list in each trial's archived requests, checks every earlier list as a prefix, and reconciles attempt counts with the final trial result and remaining-slot field. It also accounts for the 48 transfer trials. No archived submitted program is executed. This procedure avoids counting the same error again each time it appears in a later prompt.

The reconstructed tables and their crossing identities remain checkable. However, the observed scarcity and row structure occurred under this imperfect interface. A corrected description could change analysis use, record structure or actions; no such behavioral effect has been measured. Local checks of dictionary access and built-in availability cannot answer that causal question. A prospective comparison must preserve old records and compare the two interface descriptions under a separately specified allocation and budget. Mandatory reporting and external execution would still be a combined intervention in either interface arm.

Pilot sensitivity reader boundary. `artifact/pilot_sensitivity.py` regenerates all twelve single-candidate plans and their five selection metrics from the disclosed component inputs, then reconstructs the five selected indices and native-action compatibility. The corresponding P/L outcomes are linked to the archived sensitivity grades; this added check does not rerun those physical graders. It therefore closes the plan-to-selector link while retaining the numerical and historical-runtime limits of the grade evidence.

J Follow-up contract, accounting and verification boundary

The follow-up profiles are gpt-6-astra/max/Fast and gpt-6.1-sol/max/Fast. Each has 24 original assignments, eight per arm and eight matched triplets. The same four historical contexts are used, with two repeats per profile; neither repeats nor individual requests are new independent task samples. A and B execute a legal proposal. A may omit its record; an invalid optional record does not invalidate an otherwise legal proposal. B and C require a valid record and legal proposal; a protocol-invalid final produces no action. C executes the declared record selector and retains the proposal. The common schema checks case/menu identity, dimensions, finite values and legal actions. Single rows, duplicates and same-action rules are allowed. The shared analysis interface uses dictionary access and states its available operations, addressing the earlier interface mismatch; this is not a randomized old/new interface comparison, so its causal effect is unmeasured.

Each trial permits at most three optional analyses and one final decision, four model requests, 900 seconds per request and 3,780 seconds per trial. Errors consume a slot; model responses are not repaired and failed assignments are not replaced. The closed results record 36 analyses for Astra and 36 for Sol, with zero recorded analysis errors in each. These counts do not turn the historical interface comparison into a controlled experiment. Decisions were bound before privileged label lookup. Follow-up labels reuse the same identity-bound finite menu census, retaining the continuous numerical and historical-runtime premises of the earlier paper. No new physics or live crossed model decision was generated for this analysis.

Three execution segments and missingness. The original 48-position allocation was retained across three separately authorized execution segments. Positions 1–12 completed; 13 and 14 ended with HTTP 429 technical losses in the first and second segments. The third segment completed positions 15–48 without repeating either failed position. Both technical losses are Astra, in A and C of the same matched triplet. Thus Astra has 22 completed, two technical and zero protocol losses; Sol has 24 completed and zero technical or protocol losses. All 48 assignments started and have result records; none remains active or unstarted. Completion of the 34-position last segment does not mean all 48 assignments scientifically completed. Missing pairs are excluded only from observed-pair descriptions, with their denominators visible; no missing-at-random claim is made. The final segment's 88 request receipts contain no terminal 429 diagnostic. That observation does not identify the earlier limiter or establish a permanent resolution; credit exhaustion is not inferred.

Using the frozen tariffs, measured credit equivalents are 1,336.90190 for Astra and 264.37032 for Sol. Astra's two unknown requests retain a separate 1,690 credit-equivalent reserve, based on 1,050,000 input plus 128,000 output tokens per request. These are local accounting quantities, not a provider invoice or provider token ceiling. The original 7,000 local admission bound was not increased. Measured usage and unknown reserves

Profile	Requests	Known/unknown	Input	Cached input	Output
Astra	60	58/2	1,428,745	311,168	305,022
Sol	60	60/0	1,418,971	263,424	294,997

Table 12: Follow-up accounting only. Token counts cover known usage. Cached input is included in input; reasoning output is included in output. Neither subset is added again. Requests are not independent trials.

remain separate; cache splits are measured for known receipts, while any cold-price treatment of an unknown split would be an assumption, not measured zero cache.

Schema denominators. The retained field `partial_triplets` counts planned blocks missing an A, B or C assignment; its zero value does not assert scientific completion. The reader separately reports fully completed triplets (Astra 7, Sol 8), scientifically incomplete triplets (Astra 1, Sol 0), and triplets containing technical losses (Astra 1, Sol 0). The retained field `cache_split_unknown_requests` counts unknown cache splits only among requests with known token usage: its assessed denominators are 58 for Astra and 60 for Sol. Astra's two usage-unknown requests are excluded from that count and retain their full separate reserves. Thus zero unknown cache splits is neither zero cached input nor known usage for every request. The machine-readable `artifact/followup/SCHEMA.json` defines these populations and exclusions explicitly.

Reader inputs and remaining scope. The package contains 46 exact minimal finals, including 31 original numeric tables and declared rules, plus the common/A/B/C instructions and wire schema. The reader checks trial, profile, block, case and menu identities; reparses decimal tokens as exact rationals; recomputes rule selections and row diagnostics; and reconciles decisions, commit hashes and aggregates. It checks A/B against the proposals and C against the record selector. All 46 completed labels are re-read from the bound `artifact/data/crossing.json`; none changed. Section 8 gives the source paths and executable entry point. No participant analysis or task physics is run by this source-table check.

Full archives remain local. Hashes do not establish independent laboratory replication or immutable provider-model/tier identity. The proposed population, live query-selection and representation-transfer studies were not run; the finite-acquisition companion retains its separate, unchanged sample and claims.

J.1 History-level outcome accounting

Table 13 retains the four purposeful histories and their two repeats within each profile. It describes these assignments only; neither the cells nor their repeated contexts are independent new worlds. The reader derives this table and the main-text family, transition and rule-sensitivity tables from the bound minimal finals, allocation and menu census. Technical assignments retain no scientific label. All completed follow-up labels are P or L; there are no U or protocol outcomes.

Profile	History	Family	A: P/L/T	B: P/L/T	C: P/L/T
Astra	probe-01	Continuous	1/1/0	2/0/0	1/1/0
Astra	probe-02	Continuous	2/0/0	2/0/0	1/1/0
Astra	probe-03	Route	2/0/0	2/0/0	2/0/0
Astra	probe-04	Route	1/0/1	2/0/0	1/0/1
Sol	probe-01	Continuous	1/1/0	0/2/0	2/0/0
Sol	probe-02	Continuous	2/0/0	2/0/0	1/1/0
Sol	probe-03	Route	2/0/0	2/0/0	2/0/0
Sol	probe-04	Route	2/0/0	2/0/0	2/0/0

Table 13: All follow-up history/arm outcomes; T denotes technical loss with no scientific label. Every cell has two planned assignments. All completed labels are P or L. Repeated contexts are not independent draws of new worlds; P is continuation feasibility, not measured full-task completion.